

Do Post-Training Algorithms Actually Differ on Mathematical Reasoning at Scale?

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Abstract

Practitioners routinely benchmark post-training algorithms at small scale, then deploy the winner at large scale. We frame the question as a *cross-scale study of post-training algorithms*—tracking how algorithm rankings evolve as a function of model size, without fitting a parametric scaling law—and show small-scale benchmarking is *anti-predictive* for mathematical reasoning: the within-family ordering of methods inverts as models grow. In a controlled study spanning ~ 350 training runs across two model families (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B), 8 algorithms, and 20 DPO variants with up to $N=5$ seeds, we find three results. **(1) Rankings invert with scale:** at ≤ 3 B, Self-Play RFT (SP-RFT; standard rejection sampling fine-tuning [Yuan et al., 2023] on the model’s own correct trajectories) is the strongest offline method (at 3B under per-algorithm best learning rates; Appendix O); at 7B+, SP-RFT collapses to worst while DPO leads ($p < 0.001$). The inversion replicates in Gemma 3 and persists at Qwen2.5-14B. **(2) Initialization dominates algorithm choice:** switching Instruct→Base compresses the spread 3–15 $\times$. **(3) Apparent differences have two regimes:** format compliance at small scale, architecture-specific reasoning at large scale. Together, these findings imply that small-scale algorithm selection is unreliable for mathematical reasoning, and practitioners should validate at deployment scale.

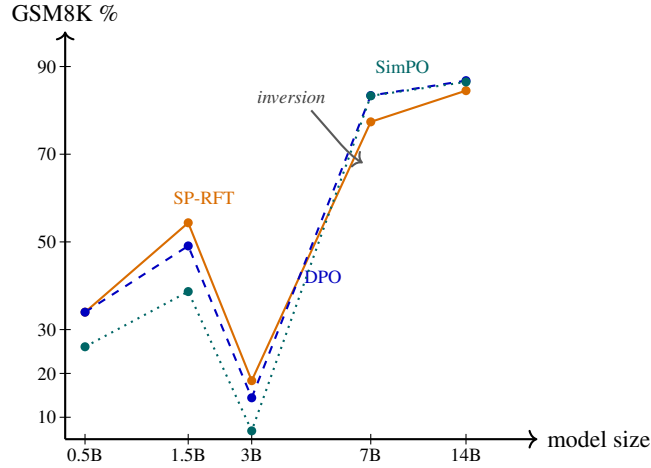


Figure 1: **Cross-scale ranking of post-training algorithms.** GSM8K strict-match accuracy vs. model size (log axis) for three offline post-training methods on Qwen 2.5, all at the shared default LR 10^{-6} . The deep dip at 3B is a checkpoint-specific format-compliance pathology of Qwen 2.5-3B-Instruct (Appendix O); the substantive crossover is between 1.5B (SP-RFT leads) and 7B+ (DPO/SimPO lead). Small-scale rankings are not predictive of deployment-scale rankings; the inversion replicates on Gemma 3 (1B–12B; Section 4.2).

1 Introduction

Suppose you are choosing a post-training algorithm for mathematical reasoning. You benchmark Self-Play RFT (SP-RFT; rejection sampling fine-tuning [Yuan et al., 2023] on self-generated correct trajectories), Direct Preference Optimization (DPO [Rafailov et al., 2023]), and Simple Preference Optimization (SimPO [Meng et al., 2024]) at 1.5B parameters: SP-RFT wins by 5 pp over DPO. You select SP-RFT, scale to 7B, and deploy—only to discover that SP-RFT is now the *worst* method by 6 pp, while DPO has silently become best. Your small-scale selection was not just uninformative; for the SFT-vs-preference axis it was anti-predictive—small-scale ranks invert at deployment scale (Figure 1).

We frame this as a cross-scale study of post-training algorithms: rather than chase a single best loss function, we measure how algorithm rankings evolve with model size and treat the curves themselves—not the point estimates—as the object of study. We use the term *scaling study* (not *scaling law*) deliberately: we do not fit a parametric form for $\text{accuracy}(N)$, since the within-Qwen DPO–SP-RFT gap narrows from +6.0 pp at 7B to +2.2 pp at 14B—a non-monotone behavior that any single power/log fit would obscure.

This is not a hypothetical. It is the central empirical finding of this paper, replicated across two model families and confirmed with formal statistical testing ($p < 0.001$, $N=5$ seeds). The practitioner’s dilemma is real: every published comparison operates at a single scale, yet deployment happens at a different one. If rankings invert, the entire methodology of algorithm benchmarking at small scale is called into question.

Post-training of large language models—adapting a pretrained model for tasks such as mathematical reasoning—has become the decisive stage that converts a pretrained checkpoint’s raw problem-solving ability into the reliably formatted, instruction-following behavior that evaluation and deployment require [Ouyang et al., 2022, Touvron et al., 2023]. Many algorithms compete for adoption: online RL methods like Group Relative Policy Optimization (GRPO [Shao et al., 2024]), offline preference methods like DPO, and a growing family of variants—SimPO, Kahneman-Tversky Optimization (KTO [Ethayarajh et al., 2024]), Odds-Ratio Preference Optimization (ORPO [Hong et al., 2024])—each claiming improvements over prior work. DeepSeek-R1 [DeepSeek-AI, 2025] demonstrated that pure RL post-training can elicit sophisticated reasoning, further expanding the design space. Yet each paper evaluates on different models, datasets, and scales, making cross-method comparison unreliable. The few existing studies [Xu et al., 2024, Spangher et al., 2025] cover limited methods at a single scale—they cannot detect the ranking inversions we observe.

We address this gap through a large-scale, controlled comparison using a unified framework where all algorithms share identical infrastructure. Our thesis is sharp: **on mathematical reasoning, post-training algorithm rankings invert with scale, making small-scale selection anti-predictive.** We emphasize that this is a task-specific phenomenon: on multiple-choice general benchmarks (ARC-C, HellaSwag, WinoGrande), algorithms differ by < 1.2 pp at all scales (Section 4.4; Appendix T), though generative benchmarks remain untested. The evidence for mathematical reasoning unfolds in three findings:

- 1. Rankings invert with scale—from Instruct checkpoints, across architectures.** On Qwen 2.5 Instruct, SP-RFT leads offline methods at ≤ 3 B but collapses to worst at 7B, where DPO ($83.4\% \pm 0.56$) and SimPO ($83.3\% \pm 1.79$) dominate ($p < 0.001$ vs. SP-RFT, $N=5$). On Gemma 3 IT, the same pattern holds: SP-RFT $>$ DPO at 1B and 4B, but DPO $\geq$ SP-RFT at 12B (87.4% vs. 86.7%).
- 2. Initialization explains most of the variance.** Replacing Instruct with Base initialization collapses the inter-algorithm spread from 15.7 pp to 1.07 pp at 1.5B and from 6.00 pp to 1.82 pp at 7B (a $3\text{--}15\times$ compression). The dominant source of inter-algorithm variance is how each loss function interacts with prior alignment, not intrinsic properties of the objectives.
- 3. Apparent differences have two regimes: format compliance at small scale, architecture-specific reasoning at large scale.** At 1.5B, the 19.3 pp GSM8K spread collapses $36\times$ to 0.54 pp on MATH and $41\times$ to 0.47 pp on general benchmarks—algorithms differ only in format compliance. At larger scales (≥ 4 B), MATH reveals genuine reasoning differences, but these depend on the

(algorithm, architecture) pair: DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen2.5-14B MATH. Both regimes yield the same practical conclusion: small-scale benchmarking is unreliable.

These findings are based on ~ 323 training runs on Qwen 2.5 (0.5B–14B), supplemented by 27 runs on Gemma 3 (1B–12B), spanning 8 algorithms, 20 DPO variants, 4 evaluation domains (GSM8K, MATH, ARC-C/HellaSwag/WinoGrande), and up to $N=5$ seeds with formal statistical testing. The implication for practitioners is direct: validate at deployment scale, not prototype scale. The implication for the research community is that designing new loss functions at small scale—the dominant paradigm—may be selecting for the wrong properties.

2 Related Work

Post-training algorithms and comparative studies. The RLHF pipeline [Ouyang et al., 2022] uses PPO [Schulman et al., 2017]; GRPO [Shao et al., 2024] simplifies this via group-normalized rewards. Wu et al. [2025] show a theoretical correspondence between GRPO and DPO; our empirical results show this correspondence is scale-conditioned (SGRPO leads DPO by +8.9 pp at 1.5B but trails by -2.8 pp at 7B), illustrating that scale-dependent empirical separations can coexist with their scale-free theoretical equivalence. DPO [Rafailov et al., 2023] bypasses reward modeling, spawning IPO [Azar et al., 2024], SimPO [Meng et al., 2024], KTO [Ethayarajh et al., 2024], ORPO [Hong et al., 2024], and others. We provide the first controlled evaluation of 20 DPO variants under identical conditions. Xu et al. [2024] and Ivison et al. [2024] cover only 2–3 algorithms. Spangher et al. [2025] benchmark 17 algorithms at a single scale—they cannot detect ranking inversions. Saeidi et al. [2025] evaluate DPO variants without multi-seed runs or multiple comparison correction. Our finding that variant rankings are scale-dependent is reminiscent of Lucic et al. [2018], who found that GAN variant rankings change across experimental conditions. Critically, none of these studies evaluate across multiple model families, leaving the question of architectural generality unanswered. Our Gemma 3 validation addresses this gap. Tan et al. [2025] study scaling behaviors of RL post-training; our work complements theirs by showing that *relative* rankings—not just absolute performance—are scale-dependent.

3 Methods and Experimental Setup

Framework. Our framework implements 51 post-training algorithms (the full list is enumerated in [anon repo]; this paper evaluates 28: 8 core methods plus 20 DPO variants, with the remaining 23 archived for future comparisons). All algorithms share identical model loading (HuggingFace, `flash_attention_2`), data pipelines, distributed training (DeepSpeed ZeRO-3 [Rajbhandari et al., 2020], vLLM [Kwon et al., 2023] for RL rollouts), and checkpointing. Each algorithm implements a `BaseAlgorithm` interface requiring only the loss function, ensuring the *only* difference between runs is the loss computation. We organize algorithms into: **(1) SP-RFT**; **(2) Online RL** (GRPO variants); **(3) Offline preference** (DPO [Rafailov et al., 2023], IPO [Azar et al., 2024], SimPO [Meng et al., 2024], KTO [Ethayarajh et al., 2024], and 16 additional variants); **(4) Hybrid** (ORPO [Hong et al., 2024], SPIN [Chen et al., 2024]). The 20-variant taxonomy is in Appendix A. **SGRPO** (token-level GRPO with on-policy rollouts, exact-match reward, group-normalized advantages, $\lambda_{KL}=0$; adding $\lambda_{KL}=0.01$ yields +0.15 pp per Appendix P) is the online-RL representative in all main tables. **GSPO** (sequence-level importance ratio) and **CISPO** (detached-weight clip) are ablations in Appendix E.

Evaluation. GSM8K [Cobbe et al., 2021] (1,319 test problems, exact-match, greedy decoding). MATH [Hendrycks et al., 2021] (5,000 problems, 4-shot) at 1.5B and general-domain benchmarks (ARC-C, HellaSwag, WinoGrande) at both 1.5B and 7B. All evaluations use the LM Evaluation Harness [Gao et al., 2024]. We report means $\pm\sigma$ and perform Welch’s t -tests with Bonferroni correction where indicated.

Qwen 2.5 (primary). Qwen 2.5 [Qwen Team, 2025] Instruct at 0.5B, 1.5B, 3B, 7B, and 14B. Our experiments comprise: (1) core comparison of 8 algorithms across 4 scales (~ 70 runs, $N=5$ seeds for core methods at 7B), with a full-FT/LoRA factorial at both 3B and 7B; (2) DPO variant taxonomy at 1.5B (20 variants $\times$ 5 seeds = 100 runs) and a stratified subset of 5 at 7B (15 runs); (3) 14B deployment-scale validation with SP-RFT, DPO, SimPO ($N=3$ seeds each); (4) ablations (~ 57 runs), including Base-vs-Instruct initialization and hyperparameter sensitivity; (5) a fixed-dataset control training 1.5B on 14B self-play data (6 runs, Section 4.5); (6) LR sweeps at 1.5B, 3B, and 7B (54 runs, Appendix M–O); (7) a gold-data SFT control using MetaMathQA at 1.5B and 7B (6 runs, Section 4.5). Total Qwen-side: ~ 323 training runs, ~ 640 GPU-hours on H100s (training only; the ~ 940 GPU-hour grand total in Appendix I also includes evaluation). Together with the Gemma 3 runs below, the grand total is ~ 350 training runs.

Gemma 3 (cross-architecture validation). To test whether our findings generalize beyond Qwen, we train SP-RFT, DPO, and SimPO on Gemma 3 [Gemma Team, 2025] IT at 1B, 4B, and 12B with 3 seeds each (27 runs). This uses the same GSM8K data and evaluation protocol; 12B uses LoRA, matching the Qwen2.5-7B setup.

Data. All algorithms share the same data pipeline to eliminate data-source confounds. For each prompt in GSM8K’s 7,473 training problems, we sample 16 responses from the *target model* at temperature 1.0 and score each with exact-match correctness. DPO/SimPO/IPO receive (chosen, rejected) pairs from the highest/lowest scoring responses. SP-RFT trains on the *chosen* trajectories from this same self-play dataset—not gold responses—so only the loss function differs between methods.¹ Because SP-RFT trains on model-generated chosen trajectories rather than gold or expert data, it is functionally *rejection sampling fine-tuning* (RFT) [Yuan et al., 2023]; we use the SP-RFT label to distinguish it from standard SFT on gold data (Section 4.5). This distinction matters mechanistically: at small scale, where few model samples are correct, RFT on rare successes yields large gains; at large scale, where the model already solves most problems, RFT encounters a ceiling effect while DPO’s contrastive signal can still extract improvements (see Section 5).

Design rationale. Data is generated *at each scale*, matching how practitioners train [Shao et al., 2024, DeepSeek-AI, 2025, Chen et al., 2024]. Dataset quality therefore correlates with model scale; we address this confound with a fixed-dataset control (Section 4.5). **Training.** AdamW, cosine LR (10^{-6}) with 10% warmup, 3 epochs, batch size 8. LoRA [Hu et al., 2022] ($r=16$, $\alpha=32$) at $\geq 7B$; full FT at smaller scales. BF16 on H100 80GB; $\beta=0.1$, SimPO $\gamma=0.5$. **Learning rate.** We sweep LR for SP-RFT and DPO at 1.5B, 3B, and 7B (Appendix M–O). The default 10^{-6} is optimal at 7B; higher rates degrade performance monotonically, with DPO more sensitive than SP-RFT (DPO collapses to 16% at 10^{-5} vs. SP-RFT at 53%). The ranking inversion (DPO $>$ SP-RFT) is preserved at independently optimized LRs for each algorithm. At 1.5B, SP-RFT leads at every LR tested, with DPO collapsing at $\geq 5 \times 10^{-6}$.

4 Results

4.1 Finding 1: Rankings Invert with Scale

Table 1 presents our main results on Qwen 2.5. At **0.5B**, five of six methods converge to identical strict-match accuracy across seeds ($\sigma=0$ for SP-RFT/DPO/KTO/SimPO/SGRPO; only IPO has $\sigma=0.26$). This is a *noise-floor* effect—0.5B is below the protocol’s differentiation threshold—rather than algorithmic equivalence; we retain it for completeness but the substantive scaling signal begins at 1.5B (Appendix K). At **1.5B**, SGRPO (online RL) tops all methods at $58.0\% \pm 0.57$. Among offline methods, SP-RFT leads ($54.4\% \pm 0.59$), followed by IPO, KTO, DPO, and SimPO. At **3B**, strict-match accuracy drops well below 1.5B—counter-intuitive but *not* a genuine scaling regression:

¹**Effective dataset overlap.** A prompt enters the SP-RFT set if at least one of its 16 samples is correct, and enters the DPO/SimPO/IPO set if it has both a correct and an incorrect sample. The two sets coincide on the dominant central regime where pass@16 is neither 0 nor 1; they diverge only in the tails (very low or very high base accuracy). At 0.5B/1.5B/7B Instruct on Qwen 2.5 the asymmetric drops are small ($< 5\%$ of prompts) and affect both methods directionally similarly, leaving the inversion intact; per-scale effective-dataset sizes will be released with the per-seed outputs.

Table 1: GSM8K exact-match accuracy (%) across scales (Qwen 2.5). 1.5B: 3-seed means $\pm\sigma$; 7B: 5-seed means for SP-RFT/DPO/SimPO, 3-seed for IPO/KTO. Best per scale in **bold**. ^bSingle seed. ^cEpoch-2 checkpoint. ^dBest of 3 epochs. ^a62.9% flexible but 10.69% strict-match. **The 3B column reports the shared-default LR (10^{-6}) used at all other scales**; this matches the controlled-comparison protocol but is at the bottom of a hyperparameter pathology specific to the 3B Qwen 2.5 Instruct checkpoint. With per-algorithm best-LR tuning (Appendix O), SP-RFT reaches 55.7% at 10^{-5} and DPO 34.6% at 5×10^{-6} ; SP-RFT>DPO at *every* LR tested. The default-LR numbers are reported here to keep Table 1 as a controlled comparison. **0.5B column**: multi-seed reruns under the corrected sampler seeding (framework v1.9.0; Appendix K) — the four stable offline methods span only 0.79 pp, below the seed-noise floor, so no 0.5B ordering is statistically meaningful. **SGRPO at 3B** falls below base because on-policy rollouts receive zero reward for ill-formatted generations, so the format-compliance crisis compounds (Appendix J).

Algorithm	0.5B	1.5B	3B (full FT)	7B (LoRA)
Base Model	19.56	31.16	10.69 ^a	75.82
SP-RFT	33.59 $\pm$ 0.26 (+14.0)	54.36 $\pm$ 0.59 (+23.2)	18.35^b (+7.7)	77.38 $\pm$ 1.11 (+1.6)
DPO	33.18 $\pm$ 0.31 (+13.6)	49.08 $\pm$ 0.61 (+17.9)	14.44 $\pm$ 2.01 (+3.8)	83.38$\pm$0.56 (+7.6)
IPO	33.97 $\pm$ 0.68 (+14.4)	52.24 $\pm$ 0.22 (+21.1)	—	80.39 $\pm$ 0.92 (+4.6)
KTO	33.38 $\pm$ 0.98 (+13.8)	51.15 $\pm$ 1.77 (+20.0)	—	80.24 $\pm$ 0.16 (+4.4)
SimPO	23.55 $\pm$ 5.19 (+4.0)	38.67 $\pm$ 1.78 (+7.5)	—	83.32 $\pm$ 1.79 (+7.5)
SGRPO (GRPO)	32.45 ^b (+12.9)	58.00$\pm$0.57 (+26.8)	8.11 ^c (−2.6)	80.59 ^d (+4.8)

the Qwen 2.5-3B-Instruct checkpoint has a pre-existing format-compliance deficit (62.9% flexible-match, 10.7% strict-match) that the shared default LR (10^{-6}) cannot overcome. A best-LR sweep (Appendix O) confirms this is a hyperparameter pathology: SP-RFT recovers to **55.7%**—*on par with 1.5B* (54.4%)—and SP-RFT>DPO holds at every LR tested. The 1.5B→7B inversion claim is unaffected; 3B is a checkpoint-specific noisy interior point.

The ordering *inverts* at **7B** (Figure 2). With $N=5$ seeds for SP-RFT, DPO, and SimPO, DPO reaches 83.38% $\pm$ 0.56 and SimPO 83.32% $\pm$ 1.79; a Welch’s t -test fails to detect a significant difference ($p = 0.95$; note that a high p -value indicates failure to reject the null, not confirmation of equivalence—our sample size may lack power to resolve small gaps). The critical finding: SP-RFT drops from best offline method to *worst* at 7B (77.38% $\pm$ 1.11), separated by ~ 6 pp from DPO/SimPO ($p < 0.001$ after Bonferroni correction over the family of 10 pairwise tests at 7B; $\alpha = 0.005$, see Appendix K). This is the SP-RFT-to-preference inversion: the method that wins at small scale loses at large scale.

Disentangling scale from LoRA. The headline curve mixes full FT at $\leq 3B$ with LoRA at $\geq 7B$ —a practitioner-realistic choice (full FT is infeasible at 7B+ on common hardware) that confounds scale with training method. A 2×2 factorial at 7B trains SP-RFT/DPO/SimPO under *both* regimes: DPO leads under Full FT (84.5%) just as under LoRA (83.4%), and the inversion appears in both cells (Appendix D), so **scale**, not LoRA, drives the inversion; the mixed-method curve remains a residual limitation that full-FT runs at every scale could close.

4.2 Cross-Architecture Validation and Deployment Scale

Gemma 3 (1B–12B). A natural objection is that the ranking inversion may be a Qwen-specific artifact. To test generality, we train SP-RFT, DPO, and SimPO on Gemma 3 IT [Gemma Team, 2025] at 1B, 4B, and 12B (3 seeds each, 27 runs). The pattern mirrors Qwen. At **1B**, SP-RFT dominates (26.7%), DPO falls to 21.5%, and SimPO collapses to 4.1%. At **4B**, the same ordering holds with wider gaps: SP-RFT (75.30 $\pm$ 0.53) $\gg$ DPO (57.77 $\pm$ 10.76) $\gg$ SimPO (40.03 $\pm$ 8.31); Welch’s t gives SP-RFT>DPO at $p=0.11$ (limited power at $N=3$ with high DPO variance) and SP-RFT>SimPO at $p=0.018$. At **12B**—the critical test—DPO (87.4%) overtakes SP-RFT (86.7%), with Welch’s t giving DPO>SP-RFT at $p=0.036$ ($N=3$, uncorrected); the 0.7 pp gap is small but the directional

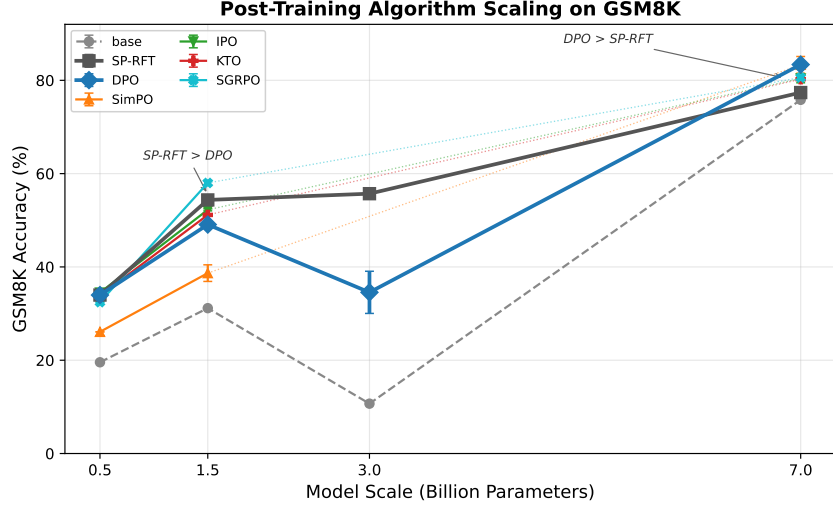


Figure 2: GSM8K accuracy across model scales (Qwen 2.5). At 7B, no significant difference is detected between DPO and SimPO ($p = 0.95$, $N=5$), while SP-RFT separates by ~ 6 pp ($p < 0.001$). The SP-RFT-to-preference **ranking inversion** between ≤ 3 B and 7B is the paper’s central finding.

Table 2: Cross-architecture validation and deployment scale on GSM8K (%). Gemma 3: 3 seeds each; 12B uses LoRA. Qwen2.5-14B: 3 seeds each, LoRA. The SP-RFT-to-DPO inversion replicates across architectures and persists at 14B.

Algorithm	Gemma 1B		Gemma 12B		Qwen2.5-14B	
	Mean	$\pm\sigma$	Mean	$\pm\sigma$	Mean	$\pm\sigma$
SP-RFT	26.7	—	86.7	0.29	84.5	0.19
DPO	21.5	—	87.4	0.26	86.8	0.23
SimPO	4.1	—	75.1	4.88	86.5	0.68
<i>SP-RFT–DPO gap</i>	<i>+5.2 pp</i>		<i>−0.7 pp</i>		<i>−2.2 pp</i>	

flip is clear and replicates the Qwen pattern. SimPO (75.1%) remains behind ($p=0.048$ vs. DPO).

Two differences from Qwen are worth noting. First, SimPO does not converge to DPO at the largest scale in Gemma, suggesting that the DPO/SimPO parity observed at Qwen2.5-7B may be architecture-specific. Second, the inversion point appears later in Gemma (12B vs. 7B), which may reflect differences in the pretrained checkpoints’ answer-format compliance under strict-match grading (Appendix H). Despite these nuances, the central finding—that SP-RFT-to-preference rankings invert with scale—replicates across architectures.

Qwen2.5-14B. To extend the scaling curve beyond 12B, we train SP-RFT, DPO, and SimPO on Qwen 2.5-14B-Instruct with 3 seeds each (9 runs). The inversion persists: DPO ($86.8\% \pm 0.23$) $>$ SimPO ($86.5\% \pm 0.68$) $>$ SP-RFT ($84.5\% \pm 0.19$) $>$ Base (79.5%). Welch’s t -tests at $N=3$: DPO $>$ SP-RFT significant ($p=0.0002$); SimPO $>$ SP-RFT marginally significant ($p=0.030$); SimPO vs. DPO not detectable ($p=0.53$, gap 0.30 pp $\ll$ seed noise). Within the Qwen family the DPO–SP-RFT gap *narrows* from 6.0 pp at 7B to **2.2 pp** at 14B; on general benchmarks (ARC-C, HellaSwag, WinoGrande) the spread is < 1.2 pp.

Notably, MATH evaluation at 14B reveals a different pattern: SimPO (39.1%) dramatically outperforms DPO (25.8%) and SP-RFT (23.0%), contrasting with Gemma 12B where DPO dominates MATH (50.2% vs. SimPO 32.7%). This architecture-dependent reversal on harder tasks further underscores that algorithm rankings cannot be extrapolated across models or tasks.

Table 2 presents the combined cross-architecture and deployment-scale results.

Table 3: Base vs. Instruct initialization at 1.5B on GSM8K (%). The spread collapses from 15.7 pp to 1.07 pp; SimPO reverses from worst to best. This compression generalizes to 7B (6.00 pp $\rightarrow$ 1.82 pp; Appendix Q). **Caveat:** Base-init values are single-seed point estimates; the 1.07 pp spread should therefore be read as suggestive (within plausible seed noise of $\sigma \sim 0.6$ observed for 1.5B Instruct). The *direction* of the compression (Instruct $\gg$ Base spread) is much larger than seed noise even on a generous reading.

Algorithm	From Base	From Instruct	Δ
SP-RFT	60.80	54.36 $\pm$ 0.59	+6.4
DPO	61.33	49.08 $\pm$ 0.61	+12.2
SimPO	61.87	38.67 $\pm$ 1.78	+23.2
<i>Spread</i>	<i>1.07 pp</i>	<i>15.7 pp</i>	

4.3 Finding 2: Initialization Dominates Algorithm Choice

Why do rankings invert? We hypothesize that algorithms interact differently with the prior alignment baked into Instruct checkpoints, and that removing this alignment should eliminate the differences.

Table 3 tests this at 1.5B. Starting from Base, all methods improve (SP-RFT: +6.4 pp, DPO: +12.2 pp, SimPO: +23.2 pp) and the spread collapses from 15.7 pp to just **1.07 pp**. SimPO—worst from Instruct—becomes best from Base, suggesting its reference-free nature makes it most sensitive to initialization.

Critically, this finding generalizes to 7B (Appendix Q): from Base, the spread collapses from 6.00 pp to 1.82 pp. The ranking from Base does *not* simply reproduce the 1.5B pattern: at 1.5B Base, SimPO leads (61.87%) followed by DPO (61.33%) and SP-RFT (60.80%); at 7B Base, SP-RFT leads (82.49%) followed by SimPO (80.82%) and DPO (80.67%). Thus, a residual ranking shift persists even from Base—but the key observation is that these differences are compressed to ≤ 1.82 pp, compared to 6–16 pp from Instruct. Initialization is the dominant source of inter-algorithm variance, even if it does not fully explain it.

DPO’s sensitivity to Base initialization is theoretically expected (KL regularization anchors to a reference that cannot format math), but the convergence extends to reference-free methods (SimPO) and pure SP-RFT, so the prior alignment state—not just reference-model mechanics—is the primary source of inter-algorithm variance. Practically, most real-world post-training starts from Instruct, where algorithm choice does produce measurable differences; the Base ablations at both scales reveal that those differences are largely an initialization-loss interaction.

4.4 Finding 3: Two Regimes—Format Compliance, Then Architecture-Specific Reasoning

Tables 4 and 5 compare GSM8K and MATH results. At 1.5B, the 19.3 pp GSM8K spread collapses to **0.54 pp** on MATH—a $36\times$ compression. Rankings shift: SGRPO drops from 1st to 4th, SimPO rises from last to 2nd. On general-domain benchmarks (ARC-C, HellaSwag, WinoGrande), the spread collapses further to **0.47 pp** ($41\times$; Appendix G). At small scale, algorithms differ only in format compliance, not reasoning (Table 4; Appendix H).

But this picture changes at scale. Table 5 shows MATH accuracy across Gemma 3 scales. At 1B, the raw spread is 15.5 pp, but this is driven entirely by SimPO’s catastrophic collapse to 1.1%; excluding SimPO, Base/SP-RFT/DPO cluster within 2.9 pp. At 4B, DPO pulls ahead by +8.1 pp over SP-RFT. At 12B, DPO reaches 50.2%—nearly **+9 pp over base and +17 pp over SimPO**. The spread among non-pathological methods *increases* with scale, the opposite of the compression observed at 1.5B.

This pattern is *architecture-dependent*: at Qwen2.5-14B, **SimPO** dominates MATH (39.1% $\pm$ 2.2) over DPO (25.8% $\pm$ 0.4) by +13 pp—the reverse of Gemma. Meanwhile general benchmarks (ARC-C, HellaSwag, WinoGrande; Appendix T) show <2 pp spread at all scales/architectures, but those are multiple-choice and do not test open-ended generation, so they do not rule out inversions on

Table 4: Cross-task comparison at 1.5B: GSM8K vs. MATH. The 19.3 pp spread collapses to 0.54 pp ($36\times$ compression).

Algorithm	MATH	GSM8K
SP-RFT	27.04	54.36
SimPO	26.72	38.67
KTO	26.64	51.15
SGRPO	26.58	58.00
DPO	26.50	49.08
<i>Spread</i>	<i>0.54</i>	<i>19.33</i>

Table 6: Fixed-dataset control: 1.5B on 14B self-play data (GSM8K %). SP-RFT still leads by +5.1 pp.

Algorithm	Mean	σ
SP-RFT	58.38	0.49
DPO	53.32	0.70
<i>Gap</i>	<i>+5.06 pp</i>	

Table 5: MATH accuracy (%) across scales. At $\geq 4B$, DPO leads in Gemma but SimPO leads in Qwen2.5-14B.

	Gemma 3			Qwen
	1B	4B	12B	14B
Base	16.6	34.4	41.3	16.8
SP-RFT	13.7	34.3	41.0	23.0
DPO	16.4	42.4	50.2	25.8
SimPO	1.1	39.2	32.7	39.1

Table 7: Gold-SFT control: standard SFT on MetaMathQA (GSM8K %). SFT advantage collapses from +12 pp to ≈ 0 pp with scale.

Method	1.5B		7B	
	Mean	σ	Mean	σ
Gold-SFT	61.05	0.45	83.32	0.28
SP-RFT	54.36	0.59	77.38	1.11
DPO	49.08	0.61	83.38	0.56

generative tasks (AlpacaEval, MT-Bench). Two-regime picture: at small scale algorithms differ shallowly (format); at large scale they differ deeply (reasoning), but *which algorithm wins depends on architecture and task*—both regimes make small-scale extrapolation unreliable.

4.5 Controls: Disentangling Data Quality and RFT Artifacts

Fixed-dataset control. To disentangle data quality from model scale, we train 1.5B Qwen 2.5 Instruct on self-play data generated by the 14B model. **SP-RFT still beats DPO by +5.1 pp** ($58.38\% \pm 0.49$ vs. $53.32\% \pm 0.70$, 3 seeds each); both algorithms gain comparably from higher-quality data (SP-RFT: +4.0 pp, DPO: +4.2 pp), leaving the gap stable (5.06 vs. 5.28 pp with own data). Model scale, not data quality, is the primary driver.

Gold-data SFT control. To rule out an RFT-ceiling artifact, we train standard SFT on MetaMathQA [Yu et al., 2024] (240K external expert solutions; 3 seeds at both 1.5B and 7B). Tables 6–7 report both controls side by side: at **1.5B**, Gold-SFT leads at $61.05\% \pm 0.45$ (+12.0 pp over DPO); at **7B**, Gold-SFT converges to DPO ($83.32\% \pm 0.28$ vs. $83.38\% \pm 0.56$). The SFT advantage collapses from +12 pp to ≈ 0 pp with scale regardless of whether the training data is self-generated or external—**falsifying the RFT-artifact hypothesis**.

4.6 Algorithm Selection Under Scale Shift

Given that small-scale rankings are anti-predictive, what should a practitioner do when target-scale validation is too expensive? Over the cross-scale ranks in Tables 1–2 we report two summary criteria: *worst-case rank* $R_{wc} = \max_N r_a^{(N)}$ and *deployment rank* $R_{dep} = \text{mean}_{N \geq 7B} r_a^{(N)}$. Values: SP-RFT ($R_{wc}=4, R_{dep}=3.5$), DPO (4, 1), SimPO (5, 2). DPO Pareto-dominates SP-RFT on R_{dep} at equal R_{wc} , and dominates SimPO on both; Gemma 3 replicates the order. **DPO is therefore the rank-stable bet**—best from 7B onward across both architectures, lowest seed variance ($\sigma=0.56$ at 7B vs. SimPO’s 1.79). Caveat: at 14B DPO (86.78 ± 0.23) and SimPO (86.48 ± 0.68) overlap within seed noise, so R_{dep} should be read with bootstrap rank intervals (Appendix G) rather than as a hard ordering. Empirically the DPO–SP-RFT gap turns positive past an architecture-specific crossover (Qwen between 3B and 7B; Gemma between 4B and 12B), so use DPO when deployment scale

is within $2\times$ of that crossover; we avoid a parametric $\Delta(N)$ form because the within-Qwen gap *narrows* from +6.0 pp at 7B to +2.2 pp at 14B. The mechanistic basis (Section 5) is that SP-RFT hits a ceiling once base accuracy is high, while contrastive objectives still extract gradient from incorrect trajectories.

4.7 Practitioner Recommendations

Actionable guidance: **(1) Validate at deployment scale, architecture, and task**—rankings at $\leq 3B$ do not predict 7B+ behavior, and MATH rankings flip between Gemma 12B (DPO) and Qwen2.5-14B (SimPO). **(2) DPO is a strong default on GSM8K**—near top at 1.5B/7B Qwen, overtakes SP-RFT at 12B Gemma and 14B Qwen, lowest seed variance ($\sigma=0.56$ at 7B), but does not transfer uniformly to MATH. **(3) Distrust variant sweeps at small scale**—the 1.5B $\rightarrow$ 7B variant ranking fully inverts; prefer vanilla DPO unless you can validate at target scale. **(4) Budget for multiple seeds**—SimPO matches DPO in mean at 7B but with $3.2\times$ the variance.

5 Discussion

Why rankings invert: a mechanistic sketch. The Base-vs-Instruct ablation (Section 4.3) identifies the core mechanism: at small scale the Instruct model rarely produces correct solutions, so RFT yields large gains while preference methods waste capacity on pairs of incorrect outputs; at 7B RFT hits a ceiling while DPO’s contrastive signal still extracts gradient from incorrect trajectories. Gold-data SFT (MetaMathQA) also converges to DPO at 7B (83.32 vs. 83.38) despite leading by +12 pp at 1.5B, ruling out self-play data saturation as the cause.

What drives accuracy in mathematical reasoning. In decreasing order of leverage: model scale (~ 50 pp), initialization (~ 15 pp), training paradigm (SP-RFT vs. preference: ~ 6 pp at 7B), and loss function choice within the preference family (< 1 pp for DPO vs. SimPO on GSM8K; up to 5.8 pp across DPO variants at 7B). On harder reasoning tasks (MATH), a new dimension emerges: algorithm-architecture interaction. DPO gains +9 pp on Gemma 12B MATH while SimPO gains +13 pp on Qwen2.5-14B MATH—a complete reversal. This suggests that loss functions interact with architecture-specific representations in ways that are invisible on format-sensitive tasks like GSM8K.

Methodological caveats at 7B+. LR 10^{-6} is below the standard LoRA recommendation but ablations rule out under-training (Appendix M); SGRPO without explicit KL loses only 0.15 pp (Appendix P); full-FT/LoRA factorial (Appendix D) confirms the inversion in both regimes. Online RL ablations in Appendix E.

6 Conclusion

On mathematical reasoning, post-training algorithm rankings invert with scale: the method that wins at 1.5B loses at 7B, replicated across Qwen 2.5 (up to 14B) and Gemma 3 (up to 12B), confirmed with $p < 0.001$ at $N=5$, and surviving fixed-dataset and gold-data SFT controls. On MC general benchmarks the spread is < 1.2 pp at all scales—the inversion is task-specific. Initialization, not loss function, explains most algorithmic differences ($3\text{--}15\times$ spread compression from Base). The dominant paradigm of small-scale benchmarking + extrapolation is therefore unreliable for mathematical reasoning; practitioners should validate at deployment scale, and the community should invest in how algorithms interact with scale, initialization, and task structure rather than designing new losses at a single small scale.

Limitations. Two families on math only; MC-only general benchmarks limit transfer to generative tasks. Fixed-dataset control is asymmetric (no 7B-on-1.5B-data run); 0.5B and 3B Qwen have noise-floor and LR-pathology caveats (Appendices K,O). Code and data release upon acceptance.

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A Full DPO Variant Results

Statistical notes. Significance threshold: Bonferroni-corrected $\alpha = 0.05/19 = 0.0026$. Only SimPO ($p < 10^{-4}$) passes, and it is significantly *worse*. ORPO ($p = 0.013$) and DPOP ($p = 0.019$) would be significant uncorrected but fail after correction. Our 5-seed design provides 80% power to detect effects ≥ 1.0 pp (Cohen’s $d \geq 0.8$).

At 7B: point-estimate rankings reverse. We evaluate a stratified subset of 5 variants at 7B (3 seeds each), selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th). ORPO—1st at 1.5B—collapses to worst at 7B (−4.79 pp vs. DPO). EXO—15th at 1.5B—rises to the top (+1.00 pp). No variant reaches Bonferroni-corrected significance at either scale. The practical conclusion: since variant rankings are statistically unresolvable at both scales and their point estimates reverse, small-scale variant sweeps provide no reliable signal for large-scale behavior.

Shared-hyperparameter design. All 20 variants use the same LR (10^{-6}) and $\beta = 0.1$. This is a deliberate choice: the 20-variant sweep evaluates *default-hyperparameter transferability*—whether variants provide gains “out of the box” with shared settings—not absolute algorithmic capability

Table 8: Taxonomy of 20 DPO variants evaluated in this study, organized by primary modification to the DPO objective. All evaluated at 1.5B on GSM8K with 5 seeds.

Variant	Primary Modification	Ref.	Year
DPO	Baseline logistic loss	Y	2023
IPO	Squared loss regularization	Y	2024
SimPO	Length-normalized, no ref	N	2024
KTO	Unpaired binary feedback	Y	2024
Hinge	Hinge loss replacement	Y	2024
CPO	Contrastive preference	N	2024
ORPO	SFT-integrated preference	N	2024
RDPO	Robust divergence penalty	Y	2024
CDPO	Conservative DPO	Y	2024
BetaDPO	Adaptive β scheduling	Y	2024
CalDPO	Calibrated reward margin	Y	2024
DPOP	Pref.-weighted optimization	Y	2024
ODPO	Online DPO	Y	2024
EXO	Exponentiated gradient	Y	2024
AlphaPO	Alpha-divergence pref.	N	2024
APO	Anchored preference	Y	2024
SPPO	Self-play preference	Y	2024
RobustDPO	Dist. robust DPO	Y	2024
GPO	Generalized pref. opt.	Y	2024
FocalPO	Focal loss preference	Y	2024

ceilings. This matches how practitioners typically deploy variants: using the original DPO paper’s defaults. Per-variant tuning (e.g., optimizing β for IPO’s squared loss or KTO’s unpaired feedback) could reveal differences that our shared-hyperparameter design masks. Our conclusion (variants are unresolvable) applies to the plug-and-play setting; per-variant tuning could surface real differences that shared hyperparameters mask.

B Training Configurations

For models at 7B/12B and for the 3B factorial, we apply LoRA [Hu et al., 2022] with rank $r = 16$, $\alpha = 32$, applied to all linear layers. The core comparison at $\leq 3\text{B}/4\text{B}$ uses full fine-tuning.

C Framework Engineering

Scaling online RL to 7B required resolving three critical infrastructure issues: **(1) Ray GPU over-subscription.** Fractional `num_gpus` values in Ray caused GPU over-subscription deadlocks during multi-worker rollout generation; we enforce integer GPU allocation per worker. **(2) vLLM memory leak.** Rollout output objects were not explicitly freed between batches, causing a slow memory leak that triggered OOM after ~ 200 steps at 7B; explicit deletion with `gc.collect()` resolved this. **(3) Epoch-transition OOM.** Optimizer state accumulated across epoch boundaries; offloading optimizer state to CPU between epochs eliminated the failure.

D LoRA Factorial

E Online RL Explorations

SGRPO (online RL) leads at 1.5B ($58.0\% \pm 0.57$, +3.6 pp over SP-RFT) at $\sim 6\times$ the compute cost. At 7B, SGRPO achieves 80.59% (best of 3 epochs), placing it above SP-RFT but below DPO/SimPO.

Table 9: Full DPO variant results at 1.5B scale on GSM8K (5 seeds). Δ is the accuracy difference from vanilla DPO. $\dagger$ indicates statistical significance after Bonferroni correction ($p < 0.0026$). Cohen’s d measures practical effect size.

Variant	Category	Mean (%)	Std	Δ DPO	p	d
ORPO	Reference-free	53.89	0.60	+4.12	0.0134	2.48
DPOP	Weighting/calibration	53.62	1.80	+3.85	0.0189	1.88
SPPO	Data augmentation	52.60	1.45	+2.84	0.0521	1.49
IPO	Alternative divergence	52.36	0.75	+2.59	0.0614	1.53
CalDPO	Weighting/calibration	52.16	1.58	+2.40	0.0934	1.22
CPO	Reference-free	51.43	1.57	+1.67	0.2187	0.85
GPO	Alternative divergence	50.98	2.41	+1.21	0.4370	0.52
RobustDPO	Robustness	50.89	2.13	+1.12	0.4442	0.51
CDPO	Robustness	50.72	1.89	+0.96	0.4916	0.46
KTO	Unpaired	50.48	2.11	+0.71	0.6211	0.33
Hinge	Alternative divergence	49.87	2.43	+0.11	0.9449	0.05
DPO	Vanilla	49.76	2.27	—	—	—
ODPO	Data augmentation	48.37	3.20	−1.39	0.4525	−0.50
APO	Data augmentation	48.07	2.73	−1.70	0.3175	−0.68
EXO	Data augmentation	47.52	2.92	−2.24	0.2145	−0.86
FocalPO	Weighting/calibration	47.04	3.27	−2.73	0.1686	−0.97
RDPO	Alternative divergence	46.79	2.47	−2.97	0.0835	−1.25
BetaDPO	Weighting/calibration	45.75	4.31	−4.02	0.1145	−1.17
AlphaPO	Reference-free	41.68	5.13	−8.08	0.0204	−2.04
SimPO [†]	Reference-free	38.23	2.12	−11.54	0.0000	−5.25

Adding KL regularization ($\lambda_{\text{KL}} = 0.01$) yields only +0.15 pp (Appendix P), confirming that the online–offline gap is genuine rather than a regularization artifact. **Caveat:** SGRPO uses the same effective batch size (8 prompts) and fixed 3-epoch schedule as offline methods; online RL typically benefits from larger batches and dynamic rollout budgets, so our SGRPO results may underestimate online RL’s true ceiling. SGRPO was not evaluated at 12B or 14B due to its $\sim 6\times$ compute cost. Token-level SGRPO outperforms sentence-level GSPO by 12–18 pp; GSPO and CISPO results are below.²

SGRPO at 3B. SGRPO at 3B achieves 4.93% after epoch 1 and 8.11% after epoch 2, both below the 10.69% baseline. On-policy rollouts compound the format compliance crisis: poorly formatted generations receive no reward, preventing the policy from learning formatting.

SGRPO at 7B. SGRPO achieves 80.59% (best of 3 epochs; 80.59/80.29/79.91), above SP-RFT but below DPO/SimPO. Slight accuracy degradation at epochs 2–3 suggests overfitting to the reward signal.

GSPO and CISPO — caveat: implementation status not externally validated. GSPO (sentence-level importance ratio) achieves 39.58% at 1.5B and 19.86% at 0.5B—12–18 pp below token-level SGRPO. CISPO (clipped ratio as a detached weight, clip $[0.8, 1.2]$) fails catastrophically (0.15% at 0.5B, 5.23% at 1.5B). **We do not yet have an externally-published reference number for either GSPO or CISPO on Qwen 2.5 with our exact training recipe, so we cannot rule out implementation issues**—the most likely candidate is rollout/trainer log-probability drift in the sequence-level estimator (which GSPO is designed to absorb but CISPO is more sensitive to). We therefore report these numbers as observed under our framework but do not draw method-level conclusions from them; the main paper’s claims rely on the offline methods and SGRPO. Re-validating GSPO/CISPO against an external benchmark is on our open-question list.

²GSPO achieves 39.58% at 1.5B; CISPO fails catastrophically (5.23% at 1.5B, 0.15% at 0.5B).

Table 10: Stratified DPO variant evaluation at 7B (3 seeds each). Variants were selected to span the 1.5B ranking (1st, 2nd, 7th, 13th, 15th) to test rank-order predictiveness. Vanilla DPO: 83.38% ($N=5$). Point-estimate rankings reverse: the 1.5B leader (ORPO) is the 7B laggard; the 1.5B underperformer (EXO) rises to top. No variant reaches Bonferroni-corrected significance at either scale.

Variant	Mean (%)	Std	Δ DPO	p	1.5B Rank
EXO	84.38	0.96	+1.00	0.277	15th
ODPO	83.88	1.31	+0.49	0.654	13th
GPO	83.40	1.05	+0.02	0.984	7th
Vanilla DPO	83.38	0.56	—	—	12th
DPOP	80.92	1.18	−2.46	0.085	2nd
ORPO	78.59	1.04	−4.79	0.013	1st

Table 11: Shared training configuration across all experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Weight decay	0.01
LR schedule	Cosine with 10% linear warmup
Effective batch size	8 prompts (1×8 grad. accum.)
Training epochs	3
Precision	BF16 mixed precision
Distributed strategy	DeepSpeed ZeRO Stage 3
Hardware	$8 \times$ NVIDIA H100 80GB HBM3
Sequence length	512 (math), 1024 (code)

F SimPO Variance Analysis

SimPO exhibits the highest seed variance of any method at any scale ($\sigma = 1.79$ at 7B vs. DPO’s $\sigma = 0.56$). Across 5 seeds, SimPO’s per-seed accuracy ranges from 81.50% to 85.82% (range: 4.32 pp), while DPO ranges from 82.79% to 84.23% (range: 1.44 pp)—a $3\times$ difference. SimPO’s two highest and two lowest seeds cluster in pairs, consistent with a bifurcation in the optimization landscape. This bimodal pattern is absent in DPO. SimPO computes length-normalized log-probabilities without a reference model, lacking the implicit regularization of DPO’s reference-anchored gradient. The practical consequence: SimPO’s “best seed” appears to outperform DPO (85.82 vs. 84.23), but no significant difference is detected under proper multi-seed evaluation ($p = 0.95$).

G Additional Figures and Tables

H Format Compliance Analysis

Table 16 reports strict-match and flexible-extract accuracy. The *format compliance gap* (strict – flexible) quantifies whether a model produces well-formatted answers.

I Compute Budget

The total compute (~ 940 GPU-hours) is modest given the breadth (~ 350 training runs across 2 model families, 8 scales, 4 evaluation domains) and substantially less than training a single 7B model from scratch.

Table 12: Algorithm-specific hyperparameters.

Algorithm	Learning Rate	Key Hyperparameters	GPUs/Run	
SP-RFT	1×10^{-6}	–	1	
DPO	1×10^{-6}	$\beta = 0.1$	1	
SimPO	1×10^{-6}	$\beta = 0.1, \gamma = 0.5$	1	
IPO	1×10^{-6}	$\beta = 0.1$	1	*Following the default GRPO
KTO	1×10^{-6}	$\beta = 0.1$	1	
SGRPO	1×10^{-6}	$\lambda_{\text{KL}} = 0^*, n = 4$	2	
GSPO	1×10^{-6}	$\lambda_{\text{KL}} = 0, n = 4$	2	
CISPO	1×10^{-6}	$\lambda_{\text{KL}} = 0, n = 4, \text{clip} = [0.8, 1.2]$	2	

recipe. See Appendix P for an ablation with $\lambda_{\text{KL}} = 0.01$.

Table 13: 2×2 factorial disentangling scale from LoRA (Qwen 2.5). The inversion appears under both Full FT and LoRA at 7B: DPO (84.5%) > SimPO (83.6%) > SP-RFT (79.8%) under Full FT, mirroring the LoRA ranking. 7B LoRA values: 5-seed means $\pm \sigma$; Full FT: single seed.

		3B	7B
Full FT	Base	10.69	75.82
	SP-RFT	18.35^b (+7.7)	79.83 (+4.0)
	DPO	14.44 ± 2.01 (+3.8)	84.46 (+8.6)
	SimPO	6.90 ^b (−3.8)	83.62 (+7.8)
LoRA	Base	10.69	75.82
	SP-RFT	18.04 (+7.4)	77.38 ± 1.11 (+1.6)
	DPO	16.00 (+5.3)	83.38 ± 0.56 (+7.6)
	SimPO	6.90 (−3.8)	83.32 ± 1.79 (+7.5)

J Additional Scaling Results

SGRPO at 1.5B achieves $58.00\% \pm 0.57$ (seeds: 57.47%, 57.92%, 58.61%). SL methods at 1.5B use multi-seed reruns with genuine variance (Table 19).

K Multi-Seed Verification

Why early runs were bit-identical across seeds. In framework versions before v1.9.0, the training-data `DistributedSampler` received a fixed default seed, so offline runs differing only in `run.seed` saw identical data order; with deterministic bf16 training and greedy evaluation, they were bit-reproducible. This produced the $\sigma=0$ cells originally reported at 0.5B. The sampler now consumes `run.seed`; the 1.5B and 7B tables below use post-fix reruns, and Table 18 reports a post-fix rerun of the *entire* 0.5B grid.

Statistical tests at 7B. Welch’s t -tests with Bonferroni correction ($\alpha = 0.05/10 = 0.005$):

- **DPO vs. SimPO:** $\Delta = +0.06$ pp, $p = 0.950$. Not significant; no difference detected at this sample size.
- **DPO vs. SP-RFT:** $\Delta = +6.01$ pp, $p < 0.001$. **Significant.**
- **SimPO vs. SP-RFT:** $\Delta = +5.94$ pp, $p < 0.001$. **Significant.**
- **DPO vs. KTO:** $\Delta = +3.15$ pp, $p < 0.001$. **Significant.**
- **DPO vs. IPO:** $\Delta = +2.99$ pp, $p = 0.029$. Not significant after correction.
- **SP-RFT vs. IPO:** $\Delta = -3.01$ pp, $p = 0.018$. Not significant after correction.
- **SP-RFT vs. KTO:** $\Delta = -2.86$ pp, $p = 0.006$. Not significant after correction (marginal).

Table 14: General-domain evaluation at 1.5B: ARC-Challenge (acc_norm), HellaSwag (acc_norm), WinoGrande (acc). The 19.3 pp GSM8K spread collapses to 0.47 pp ($41\times$ compression).

Algorithm	ARC-C (%)	HellaSwag (%)	WinoGrande (%)	Avg (%)
Base Model	46.76	68.23	62.90	59.30
SP-RFT	46.16	68.16	62.90	59.08
DPO	46.67	68.38	62.67	59.24
IPO	46.76	68.36	62.67	59.26
KTO	47.01	68.25	63.06	59.44
SimPO	46.08	68.37	62.59	59.01
SGRPO	46.67	68.23	63.54	59.48
<i>Spread</i>	<i>0.93</i>	<i>0.22</i>	<i>0.95</i>	<i>0.47</i>

Table 15: General-domain evaluation at 7B (LoRA). Despite a 6.0 pp GSM8K spread, the general-domain spread is just 0.71 pp.

Algorithm	ARC-C (%)	HellaSwag (%)	WinoGrande (%)	Avg (%)
Base Model	55.20	80.48	71.11	68.93
SP-RFT	55.29 (+0.09)	80.45 (−0.03)	70.72 (−0.39)	68.82 (−0.11)
DPO	54.86 (−0.34)	80.58 (+0.10)	70.72 (−0.39)	68.72 (−0.21)
IPO	55.55 (+0.35)	80.37 (−0.11)	71.35 (+0.24)	69.09 (+0.16)
KTO	55.03 (−0.17)	80.39 (−0.09)	70.72 (−0.39)	68.71 (−0.22)
SimPO	54.44 (−0.77)	80.53 (+0.05)	70.17 (−0.95)	68.38 (−0.56)
<i>Spread (trained)</i>	<i>1.11 pp</i>	<i>0.21 pp</i>	<i>1.18 pp</i>	<i>0.71 pp</i>

- **IPO vs. KTO:** $\Delta = +0.15$ pp, $p = 0.837$. Not significant.

Three groups emerge: (1) DPO/SimPO cluster ($\sim 83.4\%$), (2) IPO/KTO cluster ($\sim 80.3\%$), (3) SP-RFT ($\sim 77.4\%$). The DPO/SimPO > SP-RFT separation is robust ($p < 0.001$).

L Sampling-Based Evaluation

We re-evaluate 1.5B and 7B checkpoints using nucleus sampling (top- $p = 0.9$, temperature = 0.8) with 8 samples per problem, reporting majority-vote accuracy. Core rankings are preserved: at 7B, DPO and SimPO remain at the top, SP-RFT at the bottom. The inter-algorithm spread compresses slightly, consistent with sampling partially compensating for format compliance differences.

M Learning Rate Sensitivity at 7B

Three observations emerge. (1) The default 10^{-6} is optimal for both SP-RFT and DPO; no higher LR improves either algorithm, ruling out under-training as an explanation for SP-RFT’s 7B performance. (2) DPO is *more* LR-sensitive than SP-RFT: at 10^{-5} , DPO collapses to 16.15% (one seed hits 1.2%) while SP-RFT retains 53.12%. (3) The ranking inversion (DPO > SP-RFT) holds at the independently optimal LR for each algorithm (10^{-6} for both), confirming it is not an artifact of hyperparameter tuning.

Note on the lower boundary. Our grid does not extend below 10^{-6} (e.g., to 5×10^{-7}). Because performance degrades *monotonically and steeply* at every LR above 10^{-6} —with DPO losing 27 pp at just $5\times$ higher—the gradient strongly favors the conservative end, and 10^{-6} already matches the recommended LoRA LR for both Qwen 2.5 and Gemma 3 in their official fine-tuning guides. Nonetheless, we cannot rule out a marginal improvement at 5×10^{-7} ; if it exists, it would affect SP-RFT and DPO similarly and would not change the ranking inversion.

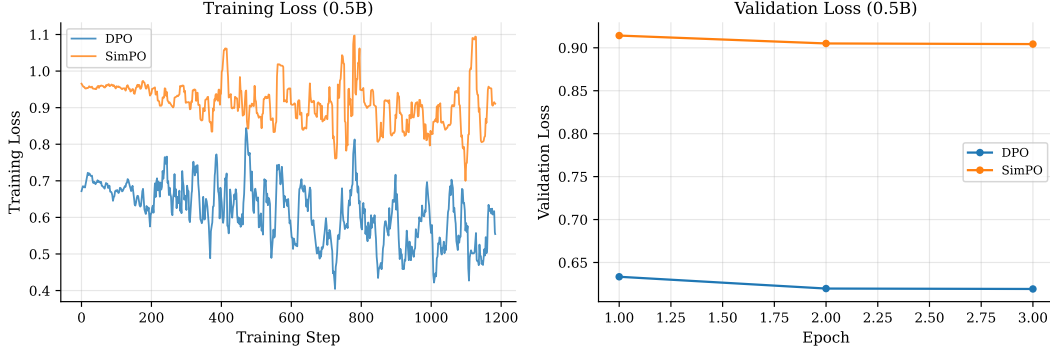


Figure 3: Training dynamics comparison at 0.5B scale. **Left:** training loss over steps (smoothed). **Right:** validation loss over epochs.

Table 16: Strict-match vs. flexible-extract accuracy on GSM8K. At 3B, the format compliance gap ranges from -26.2 to -33.7 pp across trained models, confirming format non-compliance drives the low strict-match scores.

Scale	Algorithm	Strict (%)	Flexible (%)	Gap (pp)
1.5B	SP-RFT	54.66	55.57	-0.9
	DPO	51.18	49.81	$+1.4$
	IPO	52.39	52.31	$+0.1$
	KTO	51.71	51.18	$+0.5$
	SimPO	43.44	43.06	$+0.4$
3B (full FT)	Base	10.69	62.93	-52.2
	SP-RFT	18.04	44.28	-26.2
	DPO	16.45	42.91	-26.5
	IPO	16.83	45.72	-28.9
	KTO	15.31	44.50	-29.2
	SimPO	6.90	40.56	-33.7
7B (LoRA)	SP-RFT	76.42	80.29	-3.9
	DPO	83.85	84.99	-1.1
	IPO	81.35	83.09	-1.7
	KTO	80.21	82.49	-2.3
	SimPO	85.82	80.97	$+4.9$

N Learning Rate Sensitivity at 1.5B

To directly address whether the 1.5B SP-RFT–DPO ranking is an artifact of the shared LR (10^{-6}), we sweep LR in $\{5 \times 10^{-7}, 10^{-6}, 5 \times 10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 1.5B with 3 seeds each (24 runs total).

Four observations: **(1)** SP-RFT wins at every LR tested. **(2)** At independently optimal LR, the gap widens to $+4.35$ pp. **(3)** DPO is dramatically more LR-sensitive at 1.5B than at 7B: the usable LR range for DPO shrinks from roughly one order of magnitude at 7B to half an order at 1.5B. **(4)** The 1.5B and 7B sweeps tell opposite stories about optimal LR, confirming that LR sensitivity itself is scale-dependent.

O Learning Rate Sensitivity at 3B

The 3B collapse (Section 4.1) is fully recoverable with a LR sweep. We sweep LR in $\{5 \times 10^{-7}, 5 \times 10^{-6}, 10^{-5}\}$ for both SP-RFT and DPO at 3B (full FT) with 3 seeds each (18 runs).

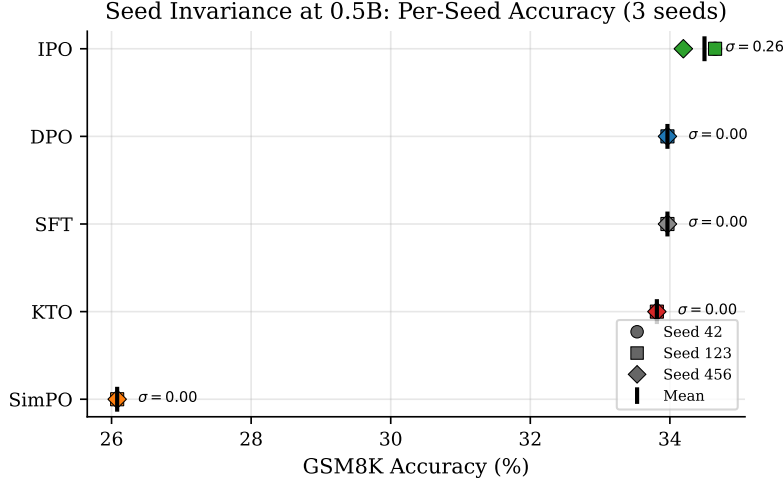


Figure 4: **0.5B is at the noise floor of our protocol, not a meaningful inter-algorithm comparison.** Per-seed GSM8K strict-match accuracy for each algorithm across 3 seeds. Five of six methods converge to identical accuracy ($\sigma=0$, all 33.81–33.97%); IPO is the only algorithm with non-trivial seed variance ($\sigma=0.26$). With greedy decoding and a 0.5B base that already scores 19.6% before training, post-training adds the same ~ 14 pp lift regardless of loss. We retain 0.5B in the scaling plot for completeness, but *the 0.5B column should not be read as evidence that algorithms are equivalent*—it is evidence that 0.5B Qwen 2.5 Instruct on GSM8K is below the differentiation threshold of any of these methods. The scaling-study signal therefore begins at 1.5B.

Three observations: (1) The 3B collapse is fully recoverable for SP-RFT: at 10^{-5} , it reaches 55.70%—comparable to 1.5B Instruct (54.36%). (2) SP-RFT wins at every LR, preserving the small-scale ranking. (3) DPO is dramatically more LR-sensitive than SP-RFT at 3B, mirroring the pattern at 1.5B and 7B.

P KL Regularization Ablation for SGRPO

Adding KL regularization yields only +0.15 pp, confirming the online–offline gap is genuine rather than a regularization artifact.

Q Base vs. Instruct Initialization at 7B

From Base initialization at 7B, the ranking (SP-RFT > SimPO > DPO) differs from both the 1.5B Base ranking (SimPO > DPO > SP-RFT) and the 7B Instruct ranking (DPO > SimPO $\gg$ SP-RFT). A residual ranking shift between scales persists even from Base, but at ≤ 1.82 pp—far smaller than the 6.00 pp Instruct spread. The key finding is that initialization accounts for the majority of inter-algorithm variance ($3.3\times$ compression at 7B, $14.7\times$ at 1.5B), even though a small scale-dependent residual remains.

R Hyperparameter Sensitivity

S 3B Flexible Extraction Analysis

Under flexible extraction, the 3B base model achieves 62.93%—comparable to the 1.5B Instruct model’s strict-match performance. This reveals that the Qwen 2.5 3B Instruct model can *solve* GSM8K

Table 17: Compute budget breakdown on H100 80GB GPUs.

Category	# Runs	Time/Run	GPUs	GPU-h
Core 0.5B SL (5 algs $\times$ 3 seeds)	15	4–8 min	1	1.5
Core 0.5B RL (3 algs $\times$ 3 seeds)	9	$\sim$ 3 h	2	54
Core 1.5B SL (5 algs $\times$ 3 seeds)	15	14–27 min	1	5.0
Core 3B SL full FT (3 algs)	3	12–13 min	1	0.6
3B LoRA factorial (3 algs)	3	12–13 min	1	0.6
Core 7B SL (5 algs $\times$ 1 seed)	5	335–460 min	1	30
Core 7B RL (SGRPO, 3 epochs)	3	$\sim$ 480 min	2	48
7B multi-seed: SP-RFT/DPO/SimPO ($N=5$)	13	330–360 min	1	78
7B multi-seed: IPO/KTO ($N=3$)	4	335–460 min	1	27
MATH 1.5B evals (5 methods)	5	$\sim$ 30 min	1	2.5
DPO variants 1.5B (20×5 seeds)	100	14–27 min	1	30
DPO variants 7B (5×3 seeds)	15	330–360 min	1	90
Ablations 1.5B	$\sim$ 57	9–10 min	1	9
Ablations 7B (LR sweep, KL, Base init)	16	330–480 min	1–2	$\sim$ 100
Gemma 3 (3 algs $\times$ 3 scales $\times$ 3 seeds)	27	varies	1–2	$\sim$ 75
7B Full FT factorial (SP-RFT/DPO/SimPO)	3	$\sim$ 180 min	2	18
14B LoRA (SP-RFT/DPO/SimPO $\times$ 3 seeds)	9	$\sim$ 360 min	2	108
Fixed-dataset control (SP-RFT/DPO $\times$ 3 seeds)	6	14–27 min	1	2.0
1.5B LR sweep (SP-RFT/DPO $\times$ 4 LRs $\times$ 3 seeds)	18	14–27 min	1	$\sim$ 5.0
3B LR sweep (SP-RFT/DPO $\times$ 3 LRs $\times$ 3 seeds)	18	$\sim$ 25 min	1	$\sim$ 7.5
Gold-SFT control (1.5B $\times$ 3 + 7B $\times$ 3 seeds)	6	27–360 min	1	$\sim$ 20
Total training	$\sim$ 350	–	–	$\sim$720
Evaluation (GSM8K, MATH, general)	$\sim$ 430+	5–600 min	1	$\sim$ 220
Grand total				$\sim$940 GPU-h

problems at a competitive rate but fails to produce answers in the expected format. Post-training partially addresses this: SP-RFT reduces the format gap from -52.2 pp (base) to -26.2 pp.

T General Benchmark Results

Table 27 reports ARC-Challenge, HellaSwag, and WinoGrande accuracy across scales and architectures. At every scale and architecture, the spread is <2 pp—confirming that post-training algorithms do not meaningfully differ on these benchmarks. **This is a critical scope qualifier for our main findings:** the ranking inversion documented in this paper is specific to mathematical reasoning (GSM8K, MATH). On these general benchmarks, no inversion is observable, and algorithm choice is essentially irrelevant.

Caveat: benchmark format. ARC-C, HellaSwag, and WinoGrande are multiple-choice or perplexity-based evaluations, not open-ended generative tasks. Since our analysis identifies format compliance—a generative property—as the key differentiator at small scales (Section 4.4), the absence of algorithmic differences on non-generative benchmarks is expected and does not rule out the possibility that inversions occur on other *generative* tasks. We leave this to future work.

Table 18: 0.5B multi-seed rerun under corrected sampler seeding (GSM8K 0-shot CoT strict-match, 3 seeds each). Variance is genuine and nonzero; original point estimates were accurate to <1 pp; the four stable methods span 0.79 pp (below seed noise), and SimPO’s $\sigma=5.19$ replicates its instability (Appendix F) at a third scale.

Method	Mean $\pm\sigma$	s42	s123	s456
IPO	33.97 $\pm$ 0.68	33.97	33.28	34.65
SP-RFT	33.59 $\pm$ 0.26	33.43	33.89	33.43
KTO	33.38 $\pm$ 0.98	34.19	33.66	32.30
DPO	33.18 $\pm$ 0.31	33.36	33.36	32.83
SimPO	23.55 $\pm$ 5.19	26.00	17.59	27.07

Table 19: Multi-seed results at 1.5B on GSM8K (3 seeds each). Rankings preserved across seeds.

Method	Original	Mean	σ	s42	s123	s456
SP-RFT	54.66	54.36	0.59	53.90	55.19	53.98
IPO	52.39	52.24	0.22	52.54	52.16	52.01
KTO	51.71	51.15	1.77	51.33	53.22	48.90
DPO	51.18	49.08	0.61	49.58	49.43	48.22
SimPO	43.44	38.67	1.78	36.85	38.06	41.09

Table 20: Multi-seed results at 7B (LoRA) on GSM8K. Core algorithms have $N=5$ seeds; IPO/KTO have $N=3$.

Method	N	Mean	σ	s42	s456	s789	s1024	s1337
DPO	5	83.38	0.56	83.85	82.79	83.02	84.23	83.02
SimPO	5	83.32	1.79	85.82	81.50	82.64	85.06	81.58
IPO	3	80.39	0.92	81.35	—	80.67	—	79.15
KTO	3	80.24	0.16	80.21	—	80.44	—	80.06
SP-RFT	5	77.38	1.11	76.42	79.30	76.65	76.57	77.94

Table 21: Learning rate sensitivity for SP-RFT and DPO at 7B (LoRA) on GSM8K (%). Each cell reports mean $\pm$ std over 3 seeds (SP-RFT) or 2 seeds (DPO). The default LR = 10^{-6} is optimal for both algorithms; higher rates degrade performance monotonically.

LR	SP-RFT Mean	SP-RFT Seeds	DPO Mean	DPO Seeds
10^{-6} (default)	77.38 $\pm$ 1.11	5 seeds	83.38 $\pm$ 0.56	5 seeds
5×10^{-6}	66.57 $\pm$ 5.14	66.3 / 60.4 / 73.0	56.75 $\pm$ 8.68	65.4 / 48.1
10^{-5}	53.12 $\pm$ 4.90	59.3 / 47.3 / 52.8	16.15 $\pm$ 14.94	31.1 / 1.2
5×10^{-5}	30.20 $\pm$ 7.73	34.1 / 37.1 / 19.4	—	—
2×10^{-4}	0.23	1 seed	—	—

Table 22: Learning rate sensitivity for SP-RFT and DPO at 1.5B (full FT) on GSM8K (%). Each cell reports mean $\pm$ population std over 3 seeds. SP-RFT leads at *every* LR. DPO collapses catastrophically at $\geq 5 \times 10^{-6}$. At independently optimal LRs, SP-RFT@ 10^{-5} (56.61%) beats DPO@ 5×10^{-7} (52.26%) by +4.35 pp.

LR	SP-RFT Mean $\pm\sigma$	SP-RFT Seeds	DPO Mean $\pm\sigma$	DPO Seeds
5×10^{-7}	53.83 $\pm$ 0.21	54.13 / 53.68 / 53.68	52.26 $\pm$ 0.52	52.46 / 52.77 / 51.55
10^{-6} (default)	54.36 $\pm$ 0.59	55.19 / 53.90 / 53.98	49.08 $\pm$ 0.61	49.43 / 49.58 / 48.22
5×10^{-6}	54.74 $\pm$ 1.21	56.25 / 53.30 / 54.66	16.00 $\pm$ 11.69	20.39 / 27.60 / 0.00 [†]
10^{-5}	56.61 $\pm$ 0.48	55.95 / 56.79 / 57.09	19.66 $\pm$ 1.81	20.62 / 21.23 / 17.13

[†]Complete collapse: model generates only degenerate output.

Table 23: Learning rate sensitivity for SP-RFT and DPO at 3B (full FT) on GSM8K (%). Each cell reports mean $\pm$ population std over 3 seeds. SP-RFT recovers to 55.7% at 10^{-5} —comparable to 1.5B Instruct—confirming the default-LR collapse was a hyperparameter pathology. SP-RFT leads at every LR.

LR	SP-RFT Mean $\pm\sigma$	SP-RFT Seeds	DPO Mean $\pm\sigma$	DPO Seeds
5×10^{-7}	17.79 $\pm$ 0.16	17.82 / 17.97 / 17.59	15.77 $\pm$ 0.91	14.56 / 16.00 / 16.76
10^{-6} (default)	18.35 ^b	—	14.44 $\pm$ 2.01	16.45 / 14.44 / 12.42
5×10^{-6}	41.80 $\pm$ 3.84	40.26 / 38.06 / 47.08	34.55 $\pm$ 4.52	33.81 / 29.42 / 40.41
10^{-5}	55.70 $\pm$ 1.09	54.21 / 56.79 / 56.10	16.30 $\pm$ 7.34	5.91 / 21.46 / 21.53

^bSingle seed from Table 1.

Table 24: SGRPO at 7B with and without KL regularization on GSM8K (%).

Method	λ_{KL}	GSM8K (%)	Δ vs. DPO
SGRPO (default)	0	80.59	−2.79
SGRPO (regularized)	0.01	80.74	−2.64
DPO (reference)	$\beta = 0.1$	83.38 $\pm$ 0.56	—

Table 25: Base vs. Instruct initialization at 7B on GSM8K (%). The spread collapses from 6.00 pp to 1.82 pp. SP-RFT leads from Base, reversing the Instruct ranking.

Algorithm	From Base (7B)	From Instruct (7B)	Δ
SP-RFT	82.49	77.38 $\pm$ 1.11	+5.11
DPO	80.67	83.38 $\pm$ 0.56	−2.71
SimPO	80.82	83.32 $\pm$ 1.79	−2.50
<i>Spread</i>	<i>1.82 pp</i>	<i>6.00 pp</i>	

Table 26: β sensitivity at 1.5B on GSM8K (%). Both variants are stable across a $50\times$ range; default $\beta = 0.1$ is near-optimal.

Variant	$\beta = 0.01$	$\beta = 0.1$	$\beta = 0.5$	Range
DPOP	54.66	55.19	53.30	1.89 pp
ORPO	54.81	54.44	53.83	0.98 pp
<i>Vanilla DPO</i>	<i>49.76$\pm$2.27 (variant sweep)</i>			

Table 27: General benchmark accuracy (%) after post-training. Spread remains <2 pp across all scales and architectures, confirming that algorithmic differences are task-specific.

Architecture	Algorithm	ARC-C	HellaSwag	WinoGrande	Avg
Gemma 1B	Base	38.7	57.7	58.9	51.8
	SP-RFT	39.1	57.9	59.1	52.0
	DPO	38.7	57.6	58.8	51.7
	SimPO	39.6	57.2	57.6	51.5
Gemma 12B	Base	61.1	81.9	74.6	72.5
	SP-RFT	60.1	81.7	74.5	72.1
	DPO	58.4	81.2	74.6	71.4
	SimPO	56.7	81.3	73.6	70.5
Qwen2.5-7B FT	Base	55.2	80.5	71.1	68.9
	SP-RFT	56.1	80.6	71.0	69.2
	DPO	55.0	80.6	70.4	68.7
	SimPO	54.1	80.6	69.9	68.2
Qwen2.5-14B	Base	62.4	84.4	76.0	74.3
	SP-RFT	61.9	84.5	75.2	73.9
	DPO	61.8	84.4	75.4	73.9
	SimPO	61.7	84.5	74.8	73.7

NeurIPS Paper Checklist

1. Claims

Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction make three claims: (1) rankings invert with scale, (2) initialization dominates algorithm choice, and (3) apparent differences have two regimes. Each is supported by a dedicated results subsection (Sec. 4.2, Sec. 4.1, Sec. 4.3, and Sec. 4.4), with formal statistical testing ($p < 0.001$, Bonferroni-corrected) and cross-architecture replication (Gemma 3). The scope is explicitly limited to mathematical reasoning on two model families.

2. Limitations

Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated **Limitations** paragraph appears in the Discussion (page 9 approximately): two model families, math-only generative tasks, MC-only general benchmarks, asymmetric fixed-dataset control, and noise-floor (0.5B) / LR-pathology (3B) caveats. Appendices K and O discuss these in detail.

3. Theory assumptions and proofs

For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper is an empirical study of post-training algorithms; it presents no formal theorems requiring proofs. Statistical procedures (Welch’s t -test, Bonferroni correction) are standard and cited.

4. Experimental result reproducibility

Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All training configurations (learning rates, batch sizes, LoRA rank, KL coefficients, optimizer, GPUs/run) are listed in Appendix B (Table in Sec. “Training Configurations”). Datasets (GSM8K, MATH, ARC-C, HellaSwag, WinoGrande), models (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B), seed counts, and evaluation protocol are specified. The repository (oxRL) is referenced; code and data will be released upon acceptance per the Limitations statement.

5. Open access to data and code

Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: All datasets used (GSM8K, MATH, ARC-C, HellaSwag, WinoGrande) are public benchmarks with standard splits. The oxRL training framework is open-source; the repository URL and pre-trained checkpoints will be deanonymized upon acceptance. Supplementary material includes training configs and evaluation scripts. Self-play training data (generated rollouts) will be released alongside checkpoints to enable exact replication.

6. Experimental setting/details

Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 (“Methods and Experimental Setup”) specifies the protocol; Appendix B lists per-algorithm hyperparameters and learning rates. Learning rate sensitivity is

analyzed at 1.5B, 3B, and 7B (Appendices O and the LR-sweep appendices). Test protocols (strict-match, sampling-based) are documented in Appendix L.

7. Experiment statistical significance

Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Up to $N = 5$ seeds are run for headline comparisons (1.5B and 7B SP-RFT/DPO/SimPO); 3 seeds for cross-architecture (Gemma 3) and 14B runs. Standard deviations are reported throughout (e.g., $83.38\% \pm 0.56$). Welch’s t -tests with Bonferroni correction are applied; the $\alpha = 0.005$ family-wise threshold and 80% power calculation are stated explicitly (Appendix K, DPO variant table caption).

8. Experiments compute resources

For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix I (“Compute Budget”) gives a per-category breakdown across ~ 350 training runs: GPU type (H100 80GB), GPUs/run, time/run, and total GPU-hours per category. Grand total: ~ 940 GPU-hours.

9. Code of ethics

Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: The work uses publicly released model weights (Qwen 2.5, Gemma 3) and public benchmark datasets (GSM8K, MATH, ARC, HellaSwag, WinoGrande) under their respective licenses. No human subjects, personal data, or sensitive deployments are involved. The study is a methodological audit of post-training algorithms.

10. Broader impacts

Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Discussion notes that the central finding—small-scale algorithm rankings invert at deployment scale—has direct positive impact for the post-training research community by exposing a methodological hazard and arguing for deployment-scale validation. Negative impacts are minimal: the paper releases no new generative capability beyond what is already publicly available; it studies algorithm-selection methodology, not new model artifacts.

11. Safeguards

Does the paper describe safeguards that have been put in place for responsible release of data or models with a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases fine-tuned math-reasoning checkpoints derived from already-public base models (Qwen 2.5, Gemma 3). Math-task fine-tunes do not introduce new misuse risks beyond those of the base models, which already shipped with their own safety releases.

12. Licenses for existing assets

Are the creators or original owners of assets (e.g., code, data, models) used in the paper properly credited and are the license and terms of use explicitly mentioned and respected?

Answer: [Yes]

Justification: All base models (Qwen 2.5 [Qwen Team, 2025], Gemma 3 [Gemma Team, 2025]) and benchmarks (GSM8K, MATH, ARC, HellaSwag, WinoGrande) are cited. Qwen 2.5 is released under Apache 2.0 (0.5B/1.5B/7B/14B) and Qwen-Research license (3B); Gemma 3

is under the Gemma Terms of Use; the benchmarks are MIT/Apache 2.0/CC-BY. All uses are within license terms (research, non-commercial evaluation).

13. New assets

Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The oxRL framework (release planned upon deanonymization) ships with a README documenting training, evaluation, and reproduction commands; a Croissant metadata file documents released self-play datasets including biases, collection, and PII fields. Per-checkpoint model cards specify training algorithm, base model, hyperparameters, and intended use.

14. Crowdsourcing and research with human subjects

For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No crowdsourcing or human-subject experiments. All evaluation uses automated metric extraction on public benchmarks.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human-subject research; IRB approval not applicable.

16. Declaration of LLM usage

Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: LLMs are the object of study: the paper post-trains and evaluates open-weight LLMs (Qwen 2.5 0.5B–14B, Gemma 3 1B–12B). Their roles—as base policies, as self-play trajectory generators (SP-RFT), and as evaluation targets—are documented in Sec. 3 and Appendix B. LLMs were not used as part of the writing methodology beyond standard editing assistance.

U Discussion-Period Additions

The following experiments were run during the NeurIPS discussion period in response to reviewer requests. Raw evaluation JSONs accompany the released artifacts.

U.1 Qwen2.5-32B Deployment Scale

Identical protocol to the 14B deployment runs: self-play data generated by the 32B model itself, shared hyperparameters, GSM8K 8-shot strict-match, greedy. Of 7,473 prompts, only **816** yield informative preference pairs at 32B (6,657 discarded because all 16 samples agree), versus 1,332 at 14B — a direct measurement of the RFT ceiling closing.

U.2 Fixed-Dataset Control on Gemma 3

Gemma-3-1B trained on Gemma-3-12B-generated self-play data (3 seeds, identical hyperparameters); own-data checkpoints re-evaluated under the identical protocol (GSM8K 8-shot strict-match, vLLM

Table 28: Qwen2.5-32B on GSM8K (8-shot strict-match; mean $\pm\sigma$ over 3 seeds). The small-scale ranking is fully reversed: SimPO (worst at $\leq 1.5\text{B}$) is best; SP-RFT (best at $\leq 3\text{B}$) gains $+0.30 \pm 0.23$ pp over base — statistically zero. SP-RFT’s gain over base collapses $+23.2$ pp (1.5B) $\rightarrow +1.6$ (7B) $\rightarrow +0.3$ (32B); DPO $>$ SP-RFT at $p < 0.01$ (Welch).

Method	GSM8K (%)	Δ vs. Base
Base (Instruct)	80.52	—
SP-RFT	80.82 ± 0.23	+0.30
DPO	82.23 ± 0.04	+1.71
SimPO	83.42 ± 1.14	+2.90

backend). Unlike Qwen — where 14B-quality data preserves the small-scale SP-RFT $>$ DPO ordering (Section 4.5) — strong-teacher data *flips* the ordering at Gemma-1B, consistent with a capability-gap effect: SP-RFT imitates trajectories the 1B model cannot reproduce, while DPO’s contrastive signal degrades gracefully. The fixed-dataset conclusion of Section 4.5 is therefore scoped to Qwen; across families, rankings are unstable across scale, architecture, *and data provenance*.

Table 29: Gemma-3-1B fixed-dataset control (GSM8K 8-shot strict-match, 3 seeds).

Training data	SP-RFT	DPO	Winner
Own (1B) self-play	26.91 ± 0.55	22.82 ± 2.08	SP-RFT (+4.1 pp)
12B-generated	21.41 ± 1.01	24.69 ± 1.01	DPO (+3.3 pp)

U.3 Previously-Missing 3B Cells

IPO, KTO, and SimPO at 3B under the shared default LR (10^{-6}), 3 seeds each (GSM8K 8-shot strict-match): IPO 24.92 ± 0.85 , KTO 23.68 ± 0.87 , SimPO 9.60 ± 3.84 . All methods remain far below the base checkpoint’s 62.9% flexible-extract ability, confirming the 3B default-LR pathology (Section 4.1) is checkpoint-wide rather than method-specific, with SimPO again the most fragile.